

Efe Onaran

CONTACT INFORMATION	David Rittenhouse Laboratory, Office: 4N55 209 S 33rd St, Philadelphia, PA 19104	sites.google.com/view/efeonaran eonaran@seas.upenn.edu
RESEARCH INTERESTS	Probability theory with a focus on the topology and geometry of random high-dimensional objects, and random growth models.	
ACADEMIC APPOINTMENTS	University of Pennsylvania 2024–present - Visiting Scholar/Postdoctoral Research Fellow - Joint appointment at the Department of Mathematics and the Department of Electrical and Systems Engineering - Mentors: Robin Pemantle, Yuliy Baryshnikov, Daniel E. Koditschek University of Illinois Urbana-Champaign 2023–2024 - Postdoctoral Research Fellow at the Coordinated Science Laboratory - Mentor: Yuliy Baryshnikov Technion – Israel Institute of Technology 2020–2023 - Postdoctoral Research Fellow - Mentors: Omer Bobrowski, Robert J. Adler	
EDUCATION	New York University 2015–2020 - Ph.D. in Electrical Engineering - Thesis title: Recovery of Permutational Data: Statistical Limits, Efficient Algorithms and Applications - Advisor: Sundeep Rangan Princeton University 2013–2015 - M.Sc. in Electrical Engineering - Thesis title: Stochastic Block Model with Side Information - Advisor: Emmanuel Abbe Bilkent University 2008–2013 - B.Sc. in Electrical Engineering	
JOURNAL PUBLICATIONS & PREPRINTS	Yuliy Baryshnikov and Efe Onaran. Ising disks: topology preserving glauher dynamics. <i>J. Appl. Comput. Topol.</i> , 10(3):Paper No. 16, 2026 Efe Onaran, Omer Bobrowski, and Robert J. Adler. Central limit theorems for local functionals of dynamic point processes. <i>Electronic Journal of Probability</i> , 31, 2026 Efe Onaran, Omer Bobrowski, and Robert J Adler. Functional central limit theorems for local statistics of spatial birth–death processes in the thermodynamic regime. <i>The Annals of Applied Probability</i> , 33(5):3958–3986, 2023 Efe Onaran and Soledad Villar. Shuffled linear regression through graduated convex relaxation. <i>arXiv:2209.15608</i> , 2022 Jinfeng Du, Efe Onaran, Dmitry Chizhik, Sivarama Venkatesan, and Reinaldo A Valenzuela. Gbps user rates using mmwave relayed backhaul with high-gain antennas. <i>IEEE Journal on Selected Areas in Communications</i> , 35:1363–1372, 2017 Efe Onaran, M Cengiz Onbasli, Alper Yesilyurt, Hyun Yong Yu, Ammar M Nayfeh, and Ali K Okyay. Silicon-germanium multi-quantum well photodetectors in the near infrared. <i>Optics express</i> , 20:7608–7615, 2012	

IN PREPARATION	Yuliy Baryshnikov, Daniel E. Koditschek, and Efe Onaran. First passage percolation with path constraints	
IN CONFERENCE PROCEEDINGS	Efe Onaran and Soledad Villar. Efficient belief propagation for graph matching. In <i>ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing</i> , pages 9060–9064. IEEE, 2020	
	Efe Onaran and Soledad Villar. Projected power iteration for network alignment. In <i>Wavelets and Sparsity XVII</i> , page 103941C. International Society for Optics and Photonics, 2017	
	Efe Onaran, Siddharth Garg, and Elza Erkip. Optimal de-anonymization in random graphs with community structure. In <i>2016 50th Asilomar Conference on Signals, Systems and Computers</i> , pages 709–713. IEEE, 2016	
	Efe Onaran, Marios Gatzianas, and Christina Fragouli. Broadcast erasure channel with feedback: The two multicast case—algorithms and bounds. In <i>2013 International Symposium on Network Coding (NetCod)</i> , pages 1–6. IEEE, 2013	
TALKS	Penn/Temple Probability Seminar	10/2024
	Bernoulli-IMS 11th World Congress in Probability and Statistics in Bochum	08/2024
	CRM-PIMS Montreal Summer School in Probability – Short talk	07/2024
	Koç University, Math Seminar – Online	04/2024
	University of Illinois Urbana-Champaign, Probability Seminar	10/2023
	Queen Mary University of London, Probability and Applications Seminar	12/2022
	Random Structures and Algorithms in Gniezno, Poland	08/2022
	Technion, Probability Seminar	04/2022
	Purdue University, Probability Seminar	03/2022
	International Conference on Acoustics, Speech, and Signal Processing – Online	05/2020
	Asilomar Conference on Signals, Systems and Computers	09/2016
	Sarnoff Symposium – Poster session	08/2016
TEACHING	University of Pennsylvania	
	• Instructor for Mathematics of Change - Part II (MATH 1080)	Fall 2025
	• Co-instructor for Advanced Applied Probability (MATH 5460)	Spring 2025
	• Co-instructor for Mathematics of Change - Part II (MATH 1080)	Fall 2024
	University of Illinois Urbana-Champaign	
	• Instructor for Optimum Control Theory (ECE 553 - MATH 595)	Spring 2024
	New York University	
	• Head TA for Applied Statistics (Undergraduate)	Spring 2020
	• Head TA for Probability Theory (Undergraduate)	Fall 2019
	• Head TA for Probability and Stochastic Processes (Graduate)	Spring 2016
	• Head TA for Signal Processing (Undergraduate)	Fall 2015
	• Grader for classes: Machine Learning, Optimization and Computational Linear Algebra, Inference and Representation, Mathematical Tools for Data Science	
SERVICE AND OUTREACH	Reviewer for	
	• Annals of Applied Probability, Electronic Journal of Probability, Statistics and Probability Letters, Linear Algebra and its Applications, IEEE/ACM Transactions on Networking, Discrete Applied Mathematics, International Symposium on Information Theory	
	Tutoring	
	• High school math tutoring for Regents Exams, Opportunity Charter School, NY	2018

HONORS AND AWARDS	Weber Fellowship at NYU Tandon School of Engineering	2015-2016
	Princeton University Department of Electrical Engineering Graduate Fellowship	2013-2014
	Highest CGPA (3.97/4.00) among all 2013 graduates of Bilkent U., School of Eng.	2013
	Full scholarship at Bilkent University	2008-2013
	Prime Ministry of Turkey Scholarship	2008-2013
CITIZENSHIP	Türkiye, United States	